Unix for Software Developers

Organizational Information about this Lecture

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Overview

1. Organizational Information
   - About the Lecture
   - About the Lab
   - About the Exam
   - Lecture Goals
2. About myself

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Organizational structure of this lecture

- Weekly lecture on Monday at **14:15 - 15:45** in **D14/2.03**
- Practical lecture, lots of examples to try out on live systems
  - Students study the provided material between lectures
  - Lecture: answers questions, do small exercises, demos
  - **I will *not* go through the material again** in classic presentation style
  - Be curious, not shy - there is a lot to discover and learn
  - Ask questions:
    - during the lecture
    - any time via email
    - to the whole course via the Moodle forum
    - Come to my office hours (TBD)

Link to the Moodle course:

`https://lernen.h-da.de/course/view.php?id=8785`

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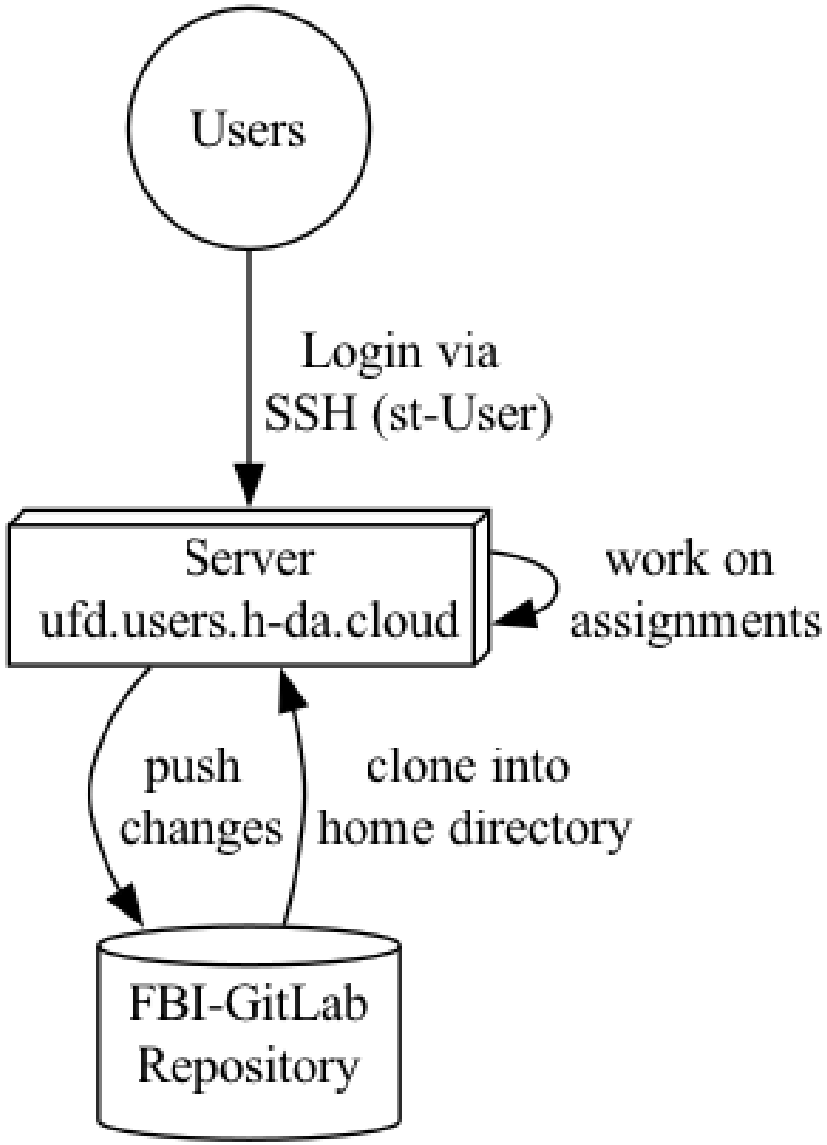
Lab organization

- Assignments posted in Moodle
- Lab attendance is **mandatory**, especially the first one!
  - If you miss a lab, you're out and someone else takes your place
  - When you get sick, go to the doctor and get a medical certificate
  - No switching groups (x or y) during the semester, the one assigned counts
- Use lab times to work on assignments and ask me questions
- **Important:** No group work, single assignment submissions
- I provide individual feedback and help during the labs
- Lab time is free work time: leave when you want/need to *after* showing me your results
- Submit solutions *before* the deadline by committing your results to GitLab
- If I find matching solutions, it counts as cheating...
- Completion of **all 5 labs** is required to take the exam
- Prepare the lab exercises at home, or you will not finish before the deadline!
- Pro tip: start early, work on them in small batches every day, ask me when you get stuck

About the Lab Assignments

- You'll get access to a GitLab project to submit your results
- A server is provided to work on the assignment solutions
- Connect to the server via SSH (see Lab 1) from your own machine
  - From University network: direct access
  - From outside: only via `h_da` VPN
  - Same goes for GitLab/git access
- From your home directory on the server commit and push your lab results
- Submit solution and lab documentation until your lab date
- Show me your solution during the lab (mandatory)
- This way, we'll get to know Unix, source control, and working in a shared environment
- Overall goal: Learn to work in the terminal, no graphical UI provided or necessary

Lab Infrastructure Diagram



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About the Exam

- Pen and paper, no practical exam
- Requires all labs marked as complete for exam participation
- German students may answer in English or German, exchange students *must* use English
- You must register separately for the exam
- Contents of the exam are the *important* parts of this lecture
- Exercises done during the lectures and labs could reappear in the exam in a different form...
- Hints:
  - Commands and sketches done during the lecture are worth writing down
  - Developers are problem solvers and that's what you'll do in the exam
  - Memorizing the lecture script will not help much
  - Do not learn command outputs as those can change
  - Knowing the basic ideas of what a Unix program is for and what it does is much more helpful

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Goals of this lecture

The lecture "Unix for Software Developers" has the following goals:

- Get to know Unix and its concepts
- Unix history is computer history
- Knowledge of Unix operating systems helps you understand different concepts in other IT areas
- Know the workflow, tools and utilities of Unix for software developers and how to use them
- Give you the help to help yourself
  - where do I find what in the system?
  - how can I solve a certain kind of problem?
- Experience the fun working with the system...

Contents and Interactions

Unix is a big topic area, impossible to show you everything that is interesting in one course. I try to give you all important topics and aim for width (a lot of topics, but not all too deep). This way, you will get a wide view about Unix topics, in which you can go deeper by yourself when interested in learning more about it.

Should I teach topics that you already know from other lectures, then tell me. Enough other topics exist that we can talk about instead of repeat topics you already know.
I try to accommodate your topic wishes, but please give me enough time to prepare them. You can ask me anytime during or after the lecture. This shows me your interest and what is not clear to you. I try to avoid lectures that turn into monologues. I would rather have a lively interaction with you, asking questions or discuss ideas. This is valuable for you as well as for me and enhances the quality of lecture.

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whoami

- Studied Computer Science at University of Applied Sciences, Darmstadt (M.Sc.)
- Administrator for the Big Data Cluster of the CS department
- Committer in The FreeBSD Project since 2009
- 6 years Vice President of the FreeBSD Foundation Board of Directors
- Serving two terms on FreeBSD's elected core team[1]
- Co-Host of the weekly BSDNow.tv[2] podcast
- FreeBSD Journal editorial board member and columnist[3]
- Speaker at open source conferences around the world (talks & tutorials)

[1] https://en.wikipedia.org/wiki/FreeBSD_Core_Team
[2] https://www.bsdnow.tv
[3] https://freebsdfoundation.org/our-work/journal/

Literature and further Information 1/2


[JFF] David Diamond, Linus Torvalds
Just for Fun - The story of an accidental revolutionary
HarperCollins, 2001


[TDAIOTFOS] Marshall Kirk McKusick, George V. Neville-Neil, and Robert N. M. Watson.
The Design and Implementation of the FreeBSD Operating System 2nd Edition
Pearson Education, Boston, MA, USA, September 2014


[AF] Michael W. Lucas
Absolute FreeBSD, 3rd Edition
No Starch Press, 2018


[FDD] Joseph Kong
FreeBSD Device Drivers: A Guide for the Intrepid
No Starch Press, 2012

Literature and further Information 2/2



[CATB] Eric Steven Raymond
The Cathedral and the Bazaar
http://www.catb.org/esr/writings/homesteading/cathedral-bazaar/



[TLDP] The Linux Documentation Project
Wikis, HOWTOs, Guides
http://www.tldp.org



[FHB] The FreeBSD Documentation Project
FreeBSD Handbook
http://www.freebsd.org/doc/en_US.ISO8859-1/books/handbook/ (english)



[NHBSD] Kirk McKusick
A narrative History of BSD
http://www.youtube.com/watch?v=ds77e3a09nA